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OPTIMAL RIVER BANK PROTECTION DESIGN BY COMBINING HYDRAULIC MODELLING WITH TURBULENCE-BASED DESIGN METHODS (*)

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SUMMARY

Damages of the bank protection on the right side of the tailrace and spillway discharge channel have been identified downstream of one of Vattenfall's hydropower plant located in Northern Sweden. The bank protection on the right side of the discharge channel, downstream the stilling basin, are made of armourstone material in three different zones: D_{50} = 1.14 m along 90 m downstream of the stilling basin, D_{50} = 0.87 m along 45 m further downstream and finally D_{50} = 0.30 m along 180 m. The affected area is located more than 120 m downstream the spillway's stilling basin, in which the bank protection is composed of the smallest armourstone size. In order to identify appropriate remediation measures, a hydraulic assessment has been performed in order to investigate the flow conditions in this region. A two-dimensional hydraulic model has been developed using high resolution bathymetry from multibeam echo-soundings and LiDAR topography. The model covers approximatively 700 m of the discharge channel. As high turbulence levels are expected during high flows, turbulence was modelled using the two-equation,

Optimisation de la conception des protections de berge en enrochements en combinant modélisation hydraulique et méthodes de dimensionnement basées sur la turbulence

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depth-averaged k-epsilon model. Results showed that if the velocities in the region where damages occurred were not extremely high, they also showed an increased turbulence level compared with the center of the channel, with the highest turbulence level occurring in the vicinity of the affected area. Turbulence-based design methods, to prevent damages caused by erosion, showed that rock sizes of D_{50} between 0.60 and 0.90 m are required along the right bank in order to withstand local turbulence levels. The method used earlier for the design of the existing bank protection was relying on cross-section-averaged velocities and schematic description of expected flow conditions. Comparison of the two methods showed that multi-dimensional hydraulic and turbulence modelling combined with turbulence-based design methods can contribute to a better understanding of erosion phenomena and to optimal bank protection design.

RÉSUMÉ

Des dommages au niveau de la protection de berge en rive droite du canal de fuite collectant le débit de la centrale et de l'évacuateur de crue ont été identifiés en aval d'un des barrages hydroélectriques de Vattenfall située dans le nord de la Suède. La protection des berges du côté droit du canal de fuite, en aval du bassin de dissipation, est constituée de blocs dans trois zones différentes : D_{50} = 1,14 m sur 90 m en aval du bassin de dissipation, D_{50} = 0,87 m sur 45 m plus en aval, et enfin D_{50} = 0,30 m sur 180 m. La zone touchée est située à plus de 120 m en aval du bassin de dissipation, où la protection des berges est composée de la fraction la plus fine. Afin de déterminer les mesures correctives appropriées, une étude hydraulique a été réalisée pour caractériser les conditions d'écoulement dans cette région. Un modèle hydraulique bidimensionnel a été mis en œuvre à l'aide d'une bathymétrie à haute résolution au sondeur multifaisceaux et d'une topographie LiDAR. Le modèle couvre environ 700 m du canal de fuite en aval du barrage. Étant donné que des niveaux élevés de turbulence sont attendus dans le canal en cas de fonctionnement de l'évacuateur, la turbulence a été modélisée à l'aide du modèle k-epsilon. Les résultats ont montré que si les vitesses dans la région où les dommages se sont produits n'étaient pas extrêmement élevées, ils indiquent que le niveau de turbulence y est plus important qu'au centre du canal. Ils indiquent également que les niveaux de turbulence les plus élevés se produisent à proximité de la zone touchée. Les méthodes de dimensionnement de protection de berge en enrochements basées sur la turbulence ont montré que des tailles de roches de D_{50} comprises entre 0,60 et 0,90 m sont nécessaires le long de la rive droite pour résister aux niveaux de turbulence locaux. La méthode utilisée antérieurement lors la conception de la protection des berges existante reposait sur des vitesses moyennes de section transversale et une description schématique des conditions d'écoulement attendues. La comparaison des deux types de méthodes a montré qu'une modélisation multidimensionnelle hydraulique associée avec un modèle de turbulence approprié, combinés avec des méthodes de dimensionnement basées

sur la turbulence, peuvent contribuer à une meilleure compréhension des phénomènes d'érosion ainsi qu'à une optimisation de la conception de la protection de berge en enrochements.

1. INTRODUCTION

1.1. Presentation of the hydropower scheme

The hydropower plant and dam highlighted in this article are located in Northern Sweden and are owned by Vattenfall. The facility is composed of two embankment dams separated by a central concrete dam hosting the powerplant and the spillway. The hydropower plant was commissioned in the early 1960s.

The spillway has three radial gates discharging in three parallel chutes exiting in a stilling basin located just next to the powerplant outlet. The discharge from the powerplant and the spillway is conveyed downstream in a tailrace channel. A site plan is presented in Fig. 1.

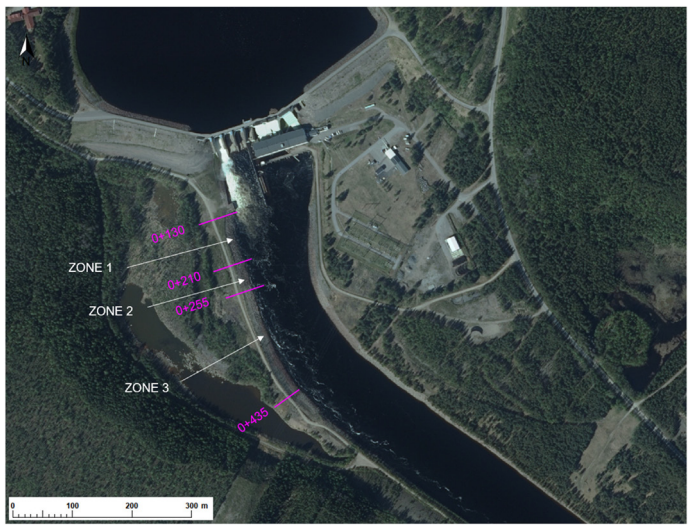


Fig. 1
*Site plan. © Swedish Land Survey authority (Lantmäteriet)
Plan de situation. © Swedish Land Survey authority (Lantmäteriet)

1.2. History of river bank protection design and damages

In 2006, reinforcement of erosion protection on both sides of the tailrace channel was done. The new bank protection was designed so that the tailrace channel could withstand a 100-year flood event without damages, while some minor damages with no consequences for dam safety could be tolerated for the facility's design flood (class I flood, which is a hydrological event that can be considered having a return period of over 10 000 years). The banks are characterized by a slope of 2H:1V.

Flow velocities and water depths determined from physical model tests were used to design the material sizes to be used in the bank protection based on the HEC-11 Design of Riprap revetment method [1]. In the following, only bank protection characteristics along the right side of the tailrace channel are considered. Three different zones have been defined, from the downstream end of the stilling basin and covering approximatively 300 m of the right bank in the downstream direction. The zones delineation and corresponding armourstone material size defined by means of the median diameter D_{50} are (see Fig. 1):

- Zone 1 – 0+130 to 0+210: D_{50} = 1.14 m
- Zone 2 – 0+210 to 0+255: D_{50} = 0.87 m
- Zone 3 – 0+255 to 0+435: D_{50} = 0.30 m

In the upstream part of Zone 1, the lower part of the bank is protected by an anchored concrete retaining wall. Only the upper part of the bank, approximatively from the normal water level, is protected by armourstone material. In the rest of the right bank, the protection is composed of two layers of armourstone material covering the whole slope and the toe.

In 2007, a discharge test was conducted for a total discharge corresponding to the 100-year flood. Damages to the newly installed bank protection were observed on both sides of the tailrace channel. The damages on the right bank occurred at the beginning of Zone 3 where the median armourstone diameter is D_{50} = 0.30 m, where maintenance repair operations were performed after the event.

Since 2007, damages to bank protection in Zone 3 have been regularly observed after the use of the spillway during spring flood, which required regular maintenance operations. Overall, bank protection damages have been observed along approximately 80 m between sections 0+250 and 0+330. The observed damages in this zone are not considered as a potential dam safety risk.

In 2022, a bathymetrical survey performed with multibeam echo-soundings revealed the presence of slumped material at the right bank toe in the upstream part of Zone 3 (between sections 0+250 and 0+300), where earlier damages were observed. Observations made during a site inspection in 2023 revealed that the

material used in the bank protection in this region have a smaller size compared with the downstream part of Zone 3 (between sections 0+330 and 0+435) where no damages were observed.

No damages have been observed in Zones 1 and 2 since 2006.

2. PURPOSE OF THE STUDY

In 2023, Vattenfall initiated a project aiming at defining remediation measures for the bank protection on the right bank of the tailrace channel. The steps involved in this project were as follows:

- Detailed hydraulic assessment in order to investigate flow conditions in the tailrace channel focused on the right bank.
- Definition of remediation measures based on the findings from the hydraulic assessment. The design criteria chosen was a 100-year flood event.

The focus of the study was put on the analysis of bank protection design in Zone 3. The rest of this article focuses on presenting the methodology and results from the detailed hydraulic assessment as well as the lessons learned regarding bank protection design.

3. HYDRAULIC ASSESSMENT

3.1. Preliminary analysis and choice of modelling approach

The analysis of historical damages along the tailrace channel's right bank showed that bank protection damages start at the limit between Zones 2 and 3, located approximatively 120 m downstream of the stilling basin and where the median diameter D_{50} of the armourstone material changes from 0.87 m down to 0.30 m (decrease by almost factor 3). This region also coincides with the beginning of a left bend of the tailrace channel. Analysis of aerial photographs showed that during spillway operation, turbulence patterns are visible in the flow all along the three bank protection zones, hinting at relatively high turbulence levels propagating downstream of the stilling basin (see Fig. 1). Hence, turbulence was identified as a key factor to be accounted for in the hydraulic assessment.

Based on data available (geometry, bathymetry) for the project, it was decided to use a two-dimensional hydraulic model (2D, depth-averaged). A 2D hydraulic model was deemed the most suitable approach being given the channel geometry that is not complex enough to require 3D or CFD modelling, while still being able to

perform high-resolution calculations using the multibeam echo-sounding survey with a low computational cost. The lack of 3D CAD model for the spillway construction was also an inconvenient for performing a throughout CFD analysis from upstream of the dam down to the tailrace channel.

3.2. Input data and hydraulic model features

The main input data to the 2D hydraulic model is a Digital Elevation Model (DEM). The DEM was created by combining the high-resolution multibeam echo-sounding survey performed in 2022 with LiDAR data for the topography above the water surface from the Swedish Land Survey authority (Lantmäteriet). Special care was given to handle the transition between LiDAR and multibeam echo-soundings in order to retain a good interpolation across the water surface in shallow areas. The DEM covers approximatively 700 m of the tailrace channel including the stilling basin and has a spatial resolution of 0.5 m. The region immediately downstream of the powerplant outlet could not be surveyed but bottom levels were recreated based on information from drawings.

The 2D hydraulic model used was TELEMAC-2D (version 8.3.0) from the open-source suite openTELEMAC-MASCARET [2]. TELEMAC-2D solves the Shallow Water Equations on an unstructured triangular mesh using the Finite Element Method with a large choice of turbulence models [3].

The 2D model developed has the same extent as the DEM. The computational mesh is composed of approximatively 250 000 triangular cells with an edge length of 1 m in the whole domain. The model domain and the bottom elevations as well as the location of the bank protection damages and slumped material identified in Zone 3 are illustrated in Fig. 2.

The model has four boundaries: the three spillway chutes (boundaries are located inside the stilling basin) where the total inflow corresponding to a 100-year flood event is prescribed, as well as the downstream boundary located approximatively 700 m downstream of the stilling basin where the stationary water level corresponding to the same flood event is set as boundary condition (+46.15 m.a.s.l.). It was considered that all the 100-year flood was discharged through the spillway with no turbine flow through the powerplant.

Turbulence was modelled using the depth-averaged k-epsilon turbulence model in order to use the computed turbulent kinetic energy as input to bank protection design. The depth-averaged k-epsilon turbulence model belongs to the family of Reynolds-averaged Navier Stokes (RANS) based turbulence models [3] [4].

As the length of the model is limited in the streamwise direction (approximatively 700 m), no significant friction losses develop between the upstream and downstream ends, making calibration in terms of bottom friction coefficient not

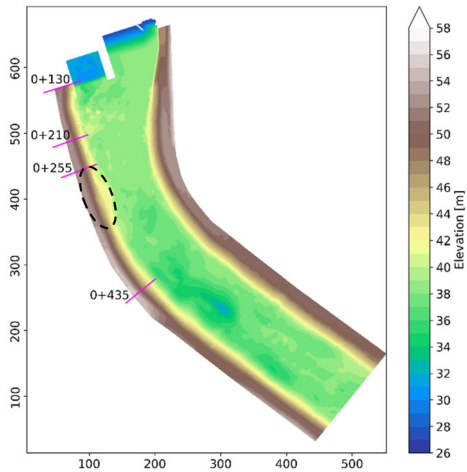


Fig. 2
Model domain and bottom elevations. Dashed ellipse: location of bank protection damages and visible slumping at bank toe
Emprise du modèle numérique et bathymétrie. Ellipse : emplacement des dommages et de l'éboulement identifié en pied de berge

relevant. Bottom friction was modelled using a Strickler coefficient of K_s = 30 m$^{1/3}$/s, which corresponds to a Manning coefficient n = 0.0333 s/m$^{1/3}$. A sensitivity analysis on this parameter has been performed in order to estimate its influence on flow conditions. No significant influence was identified, hence all the simulations were performed with the aforementioned values.

3.3. Turbulence-based erosion protection design methods

Erosion protection design methods are usually based on the incipient motion concept defined by the Shields equation. The Shields equation states that the load required to set particles into motion (called the critical bed shear stress τ_c [N/m^2]) is proportional to the strength of the bed material defined as $(\rho_s - \rho)gD_{50}$, with the proportionality factor being called the critical Shields parameter Ψ_c [-]:

$$\tau_c = \Psi_c (\rho_s - \rho) g D_{50} \quad (1)$$

With ρ_s the material specific density [kg/m^3], ρ the water density [kg/m^3], g the acceleration of gravity [m/s^2] and D_{50} the median particle diameter [m]. The value of

the critical Shields parameter Ψ_c varies in function of material type and particle Reynolds number. For natural materials (sediments to stones), Ψ_c typically varies between 0.02 and 0.06 [5].

One of the main challenges in applying incipient motion methods based on the Shields equation is the estimation of the bed shear stress. The bed shear stress is often estimated assuming steady-state flow conditions. If this assumption is usually acceptable for uniform flow conditions, it can become uncertain in non-uniform flow conditions in which the dynamic bed-pressures variations, arising from flow turbulence, can significantly affect the onset of material erosion.

The concept of turbulence-based erosion protection design methods is to explicitly account for flow turbulence in the load term used in the Shields equation. The turbulence parameter used is called the relative turbulence intensity r_0 [-] defined as [5]:

$$r_0 = \frac{\sqrt{k_0}}{U_0} \tag{2}$$

With k_0 the depth-averaged turbulent kinetic energy [m^2/s^2] and U_0 the depth-averaged flow velocity [m/s]. By combining the Shields equation (Eq. [1]) with the definition of bed shear stress τ_0 [N/m^2]:

$$\tau_0 = \rho u_*^2 \tag{3}$$

and with the relationship between the depth-averaged turbulent kinetic energy k_0 and the friction velocity u_* [m/s]:

$$\sqrt{k_0} = \alpha_0 u_* \tag{4}$$

Hoffmans proposed the following design equation [5]:

$$D_{50} = 0.7 \frac{(r_0 U_0)^2}{\Delta g \Psi_c} \tag{5}$$

With Δ the relative material density defined as $(\rho_s - \rho)/\rho$ and using a numerical value of 1.2 for the coefficient α_0 as suggested in [5].

The equation is valid for bed material in both uniform and non-uniform flow conditions [5]. In uniform flow conditions, differences between this method and the classic methods based on bed shear stress are small as flow turbulence is generated by bottom friction. On the other hand, in non-uniform flow conditions, differences between the two types of methods might arise due to the direct influence of local flow turbulence on shear stress at near the bed. Typical values for relative turbulence intensity are presented in Table 1.

Table 1
Relative turbulence intensity typical values [5]
Valeurs indicatives pour l'intensité turbulente relative [5]

r_0	**Turbulence level**	**Remarks**
0	No turbulence	Laminar flow
< 0.08	Small turbulence	-
0.08 – 0.15	Normal turbulence	Channel, river flow
0.15 – 0.20	High turbulence	Downstream of structures (bridges, piers, etc.)
0.20 – 0.30	Very high turbulence	Downstream hydraulic jumps, sharp bends, etc.
0.30 – 0.60	Extreme turbulence	-

The design equation proposed by Hoffmans (Eq. [5]) can easily be combined with 2D hydraulic models in which the k-epsilon turbulence model is used as U_0 and k_0 are variables given directly by the numerical model.

For erosion protection design, the critical Shields parameter can be used as a design criterion for permissible damage. A typical value for riprap revetments used in bank protection is Ψ_c = 0.03 while a value of Ψ_c = 0.02 represents a more conservative criterion for permissible damage [5].

For bank protection design, D_{50} calculated with Eq. [5] should be divided by the bank slope factor K_{sl} [-]:

$$K_{sl} = \sqrt{1 - \left(\frac{\sin\theta}{\sin\Phi}\right)^2} \tag{6}$$

With θ the bank slope angle and Φ the material friction angle which is usually 40º for bank protection material (armourstone made of blasted rock). For this study, θ = 26.6º yielding K_{sl} = 0.72 are used. The results regarding calculated D_{50} values presented in the following section are based on Eq. [5] which has been applied for both Ψ_c = 0.03 and Ψ_c = 0.02. For bank protection design, D_{50} should be calculated with Ψ_c = 0.03 and then been divided by K_{sl}. As 0.02/0.03 $\approx$ 0.72, the results obtained with Eq. [5] and Ψ_c = 0.02 were used as input for the bank protection design.

3.4. Results

Flow conditions in the tailrace channel for the 100-year flood event and for the current bathymetry (with the slumped material left in place at the upstream end of

Zone 3) are illustrated in terms of water depth, depth-averaged flow velocity, depth-averaged turbulent kinetic energy and relative turbulence intensity in Fig. 3.

Water depth along the right bank toe varies from approximatively 6 m in Zone 1 to 7-8 m along Zone 2 and progressively increases along Zone 3 from 7-8 m to locally approximatively 10 m. Water depth in the center of the channel follows the same pattern with a gradual increase from approximatively 6-7 m in front of Zone 1 to 10-12 m in Zone 3. Water depth in the region where damages were observed (upstream part of Zone 3) is varying between 7-8 m.

Analysis of flow velocities shows that they follow an inverse pattern compared with water depth in the central part of the channel, with the highest velocities (approximatively 4 m/s) occurring in front of Zone 1 and decreasing progressively to 3-3.5 m/s along Zone 3 before decreasing even further downstream where water depth increases. Along the channel banks, two regions with circulation zones (eddies) are observed, one along the right bank in Zone 3 starting approximatively from the location of the observed damages and one along the left bank in front of the downstream end of Zone 3. In these eddies, the steady-state velocity is lower than 1 m/s. Finally, a large eddy is also occurring in the region immediately downstream of the powerplant outlet. The presence of this eddy is linked to the assumption that no flow is conveyed through the powerplant.

Analysis of the turbulent kinetic energy and relative turbulence intensity shows that a very high turbulence level occurs in the three regions affected by eddies. The turbulent kinetic energy is high along the right bank in the downstream part of Zone 3. In the center part of the channel, the relative turbulence intensity varies from approximatively 0.5 in the stilling basin (which corresponds to the value assumed and set as boundary condition at each spillway chute), progressively decreasing to 0.2 at the limit between Zones 2 and 3 and then further decreasing towards 0.15 in the remaining part of the channel, which corresponds to an expected turbulence level (normal to high) in such a channel geometry for the flood event under consideration (see Table 1).

Overall, results show that flow conditions along Zone 3 are challenging with an eddy inducing high turbulence levels along the bank and with flow velocities larger than 3 m/s in the central part of the channel outside the eddy.

A computation made for a configuration with the slumped material removed showed only minor differences on flow conditions. The size of the eddy located along Zone 3 is slightly smaller, which implies that the flow velocities near the bank slope toe are slightly increased. The turbulence level inside the eddy remains high.

Finally, computed flow velocities and relative turbulence intensities were used in order to produce maps of corresponding median armourstone diameter based on Eq. [5] proposed by Hoffmans and for the two critical Shields parameters considered (Ψ_c = 0.03 and Ψ_c = 0.02), see section 3.3.

The two maps presented in Fig. 4 correspond to the current geometry and to the geometry in which the slumped material at the upstream end of Zone 3 has been removed. Results have also been extracted along a profile located approximatively at the toe of the bank slope. Values for velocity, relative turbulence intensity and calculated D_{50} based on the equation proposed by Hoffmans and for Ψ_c = 0.02

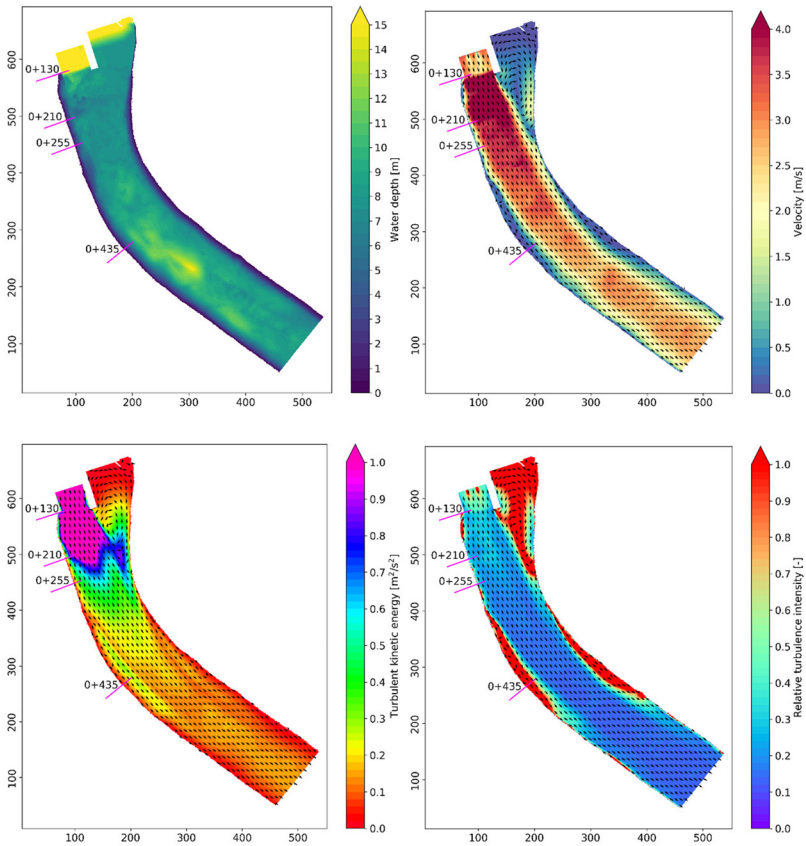


Fig. 3
Flow conditions for the 100-year flood. Top left to bottom right: water depth, depth-averaged velocity, depth-averaged turbulent kinetic energy and relative turbulence intensity with velocity vectors.
Résultats pour le débit centennal. De gauche à droite et de haut en bas : profondeur, vitesse moyenne sur la verticale, énergie turbulente et intensité turbulente relative. Les vecteurs matérialisent le champ des vitesses.

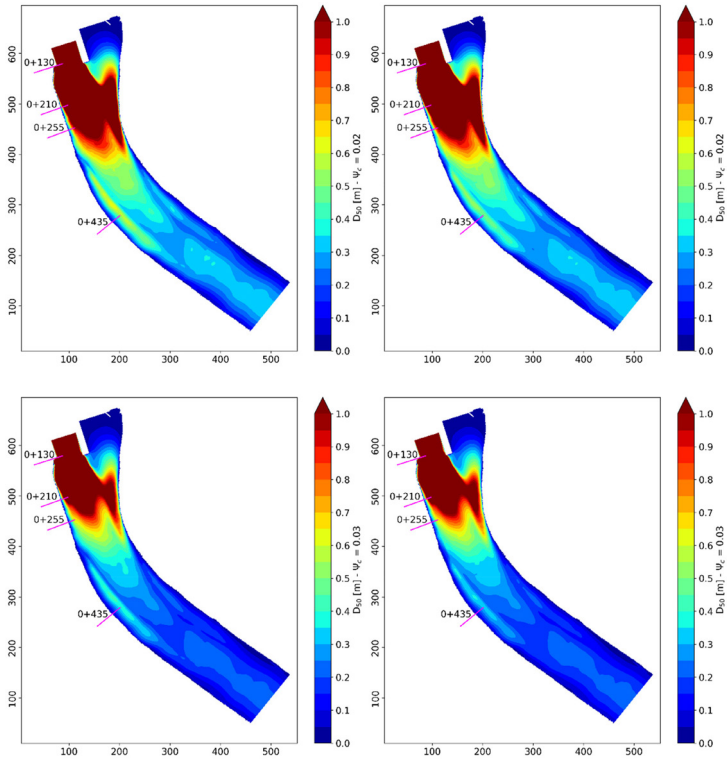


Fig. 4

Calculated D_{50} for the 100-year flood. Images on left side: current geometry with slumped material in place (between sections 0+250 and 0+300). Images of right side: geometry with slumped material removed. Upper images: Ψ_c = 0.02. Lower images: Ψ_c = 0.03

D_{50} calculé pour le débit centennal. Images de gauche : bathymétrie actuelle avec éboulement en place (entre sections 0+250 et 0+300). Images de gauche : bathymétrie sans éboulement. Images du haut : Ψ_c = 0.02. Images du bas: Ψ_c = 0.03

(value chosen for bank protection design, see section 3.3) are presented in Fig. 5 for the current geometry and for the geometry in which the slumped material at the upstream end of Zone 3 has been removed.

The flow velocity along the right bank oscillates between approximatively 4 and 5 m/s in Zone 1 and then progressively decreases to reach approximatively 3 m/s at the downstream end of Zone 2. The relative turbulence intensity in this part of the channel oscillates between approximatively 0.2 and 0.3. In Zone 3, both velocity

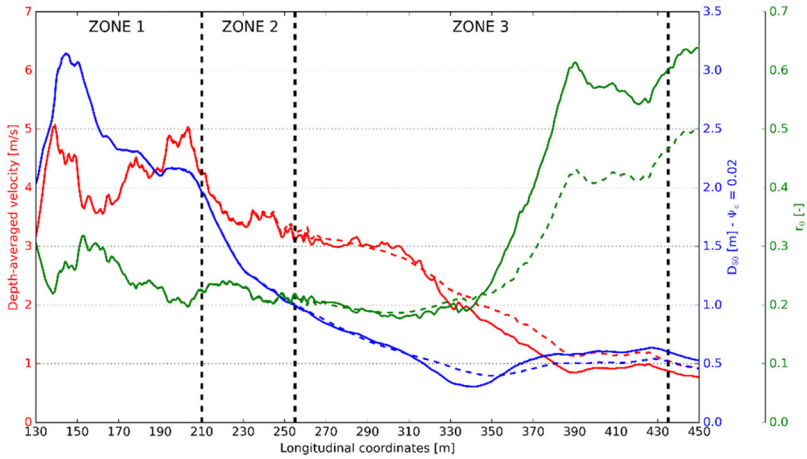


Fig. 5

Longitudinal profile along right bank extracted at bank slope's toe with depth-averaged velocity, relative turbulence intensity and calculated D_{50} for the 100-year flood. Solid lines: current geometry with slumped material in place Dashed lines: geometry with slumped material removed.

Profile en long de la rive droite en pied de berge avec vitesse moyenne sur la hauteur, intensité turbulente relative et D_{50} calculé pour le débit centennal. Lignes pleines : bathymétrie actuelle. Lignes en pointillés : bathymétrie sans éboulement.

and relative turbulence intensity remains approximatively constant (3 m/s and 0.2, respectively) until the section 0+310 that materializes the downstream part of the observed bank protection damages. Downstream of this point, flow velocity decreases progressively down to approximatively 1 m/s around section 0+390 while the relative turbulence intensity increases up to 0.4-0.6. This region corresponds to the outer part of the eddy occurring along the right bank. The comparison between results for the current geometry and for the geometry in which the slumped material at the upstream end of Zone 3 has been removed shows that velocity downstream of section 0+310 are up to 20% higher with the latter geometry while the relative turbulence intensity decreases by approximatively 50% from 0.6 down to 0.4. Although the size of the eddy and its turbulence level is larger in the presence of slumped material at the bank slope's toe, the removal of the slumped material does not allow a significant diminution of the turbulence level in this region. Hence, it is likely that the natural high turbulence level in this region contributed to the apparition of bank protection damages during the discharge test performed in 2007.

The calculated D_{50} in each zone (interval between minimum and maximum values) based on Ψ_c = 0.02 (value chosen for bank protection design, see

section 3.3) are summarized in Table 2, together with the D_{50} of the existing bank protection.

Table 2
Summary of calculated D_{50} in each zone (minimum and maximum values) for the geometry with slumped material removed for the 100-year flood and comparison with existing D_{50}
Synthèse des valeurs de D_{50} calculées dans chaque zone (minimum et maximum) dans la configuration sans éboulement et comparaison avec les D_{50} existants. Débit centennal

	Zone 1	**Zone 2**	**Zone 3**
D_{50} (Ψ_c = 0.02) [m]	2.0-3.1	1.0-2.0	0.4-1.0
Existing D_{50} [m]	1.14	0.87	0.30

Comparison shows that the calculated D_{50} is always larger than the values for the existing bank protection. The analysis of the longitudinal profile shows that the calculated D_{50} varies strongly along each zone, with the highest value always located at the upstream end of each zone (excepted for Zone 1 where the maximum values is reached after 15 m).

The comparison between the calculated and the existing D_{50} in Zones 1 and 2 should be interpreted with care. The upstream part of Zone 1 features a different design with anchored retaining wall for the lower part of the bank, thus increasing the overall protection against erosion. Furthermore, the fact that no damages have historically been observed in this region might indicate that the design method used to calculate D_{50} is potentially yielding conservative results. Finally, as this region is located near the upstream boundaries of the 2D hydraulic model and immediately downstream of the stilling basin, it is likely that the results from the model contain significant uncertainties. Further analysis with more sophisticated methods or observations is required in order to draw final conclusions regarding suitability of armourstone size in these two zones.

The analysis of the results along Zone 3, which was the focus of the study, shows that the armourstone size of the existing bank protection (D_{50} = 0.30 m) is inferior to the calculated D_{50}. At the beginning of the zone, the calculated D_{50} is approximatively three times larger than the existing D_{50}. In the second half of the zone, the calculated D_{50} is approximatively 0.50 m, which is 67% larger than the existing armourstone (D_{50} = 0.30 m).

As a conclusion, the hydraulic assessment performed in the tailrace channel in combination with a turbulence-based bank protection design method has shown that the existing armourstone size in the upstream part of Zone 3 (right bank), where flow conditions are characterized by a high turbulence level, does not fulfill design

requirements for the design flood (100-year return period). These findings are in agreement with the observation of damages that has regularly occurred in this region for discharges lower than the design flood.

3.5. Comparison with other bank protection design methods

It is interesting to compare the calculated D_{50} obtained from the combination of 2D hydraulic modelling and the design method proposed by Hoffmans (see section 3.3) with other classical bank protection design methods. A collection of bank protection design methods is available in the Rock Manual handbook [6], and the following methods have been used: Pilarczyrk, Escarameia & May as well as Maynord. Additionally, the classic Isbash method [7] as well as the HEC-11 method [1] used in the original design from 2006 were also used. Among these methods, three of them use turbulence characteristics as an explicit input parameter (relative turbulence intensity for Hoffmans and Escarameia & May and a turbulence factor for Pilarczyrk). In the other three methods (Maynord, Isbash and HEC-11), coefficients are used to account for expected flow conditions, type of construction, river sinuosity, but no explicit guidance is given to account for flow turbulence.

The methods have been applied for the flow and geometrical characteristics corresponding to the upstream part of Zone 3 along the right bank (see Fig. 5): flow depth of 7.5 m, velocity of 3.0 m/s, relative turbulence intensity of 0.2, bank slope of 2H:1V ($\approx$ 26.6º) and material friction angle of 40º. As the main part of the methods considered are based on a design Shields parameter of $\Psi_c \approx 0.03$, only this value has been used for the Hoffmans method in which the bank slope factor K_{sl} was applied (see section 3.3, Eq. [6]). For the Pilarczyrk method, a turbulence factor k_t^2 = 2.0, corresponding to “non-uniform flow, sharp outer bends”, was chosen. The HEC-11 method was applied with a flow velocity of 3.3 m/s (average velocity in the channel) and a material friction angle of 32º, which correspond to the parameters used in the original design from 2006. The results are presented in Table 3.

The comparison of the results reveals that the three methods that use turbulence characteristics as an explicit input parameter (Hoffmans, Pilarczyrk and Escarameia & May) yield the largest D_{50} values of the set, with a relatively large spread between 0.50 and 0.87 m. The Pilarczyrk method gives D_{50} values of 0.55 and 0.87 m for a fully developed and for a not fully developed velocity profile, respectively. The Hoffmans method lies in between those two values, with D_{50} = 0.72 m.

The three other methods (Maynord, Isbash and HEC-11), when used with the standard assumptions, yield D_{50} values significantly lower than the three methods mentioned above, between 0.19 and 0.30 m. However, the Isbash method applied in outer bends for a bend angle of 30º similar to the left bend in the tailrace channel,

Table 3
Calculated D_{50} with different bank protection design methods
Valeurs de D_{50} calculées avec différentes méthodes de dimensionnement

Design method	**Remarks**	D_{50} **[m]**
Hoffmans (Eq. [5])	Ψ_c = 0.03, bank slope factor applied (Eq. [6])	0.72
Pilarczyrk	Fully developed velocity profile	0.55
	Not fully developed velocity profile	0.87
Escarameia & May	-	0.50
Maynord	Straight channels, inner bends	0.19
	Downstream of concrete structures / end of dikes	0.25
	Outer bends	0.23
Isbash	Straight channels	0.22
	Outer bends (30º)	0.50
HEC-11	SF = 1.6 (original design, 2006)	0.30
	SF = 2.0	0.42

yields a D_{50} value of 0.50 m. In the original design from 2006, the HEC-11 method was used with a so-called stability factor SF = 1.6 corresponding to the upper range of flow conditions characterized by gradually varying flow and moderate bend curvature. Using SF = 2.0, corresponding to the upper range of flow conditions characterized by rapidly varying flow, sharp bend curvature, turbulently mixing flow at bridge abutments and uncertain design parameters, yields a D_{50} value of 0.42 m.

Changing the turbulence characteristics used in the Hoffmans, Pilarczyrk and Escarameia & May methods to match normal turbulence levels during floods (relative turbulence intensity of 0.10-0.15, turbulence factor k_t^2 = 1.0) yields D_{50} values in the same range as the non-explicitly turbulence-based methods with standard assumptions ($\approx$ 0.2-0.3 m), unless for Pilarczyrk with a not fully developed velocity profile that gives a slightly larger estimate (0.35 m).

The comparison presented in this section shows that turbulence-based bank protection design methods lead to larger armourstone dimensions than other design methods where flow turbulence is high. The findings from this specific study, based on observations of historical bank protection damages, suggest that turbulence-based bank protection design methods should be used when high flow turbulence is expected. This conclusion is in agreement with a similar comparison reviewed in [5].

4. DEFINITION OF REMEDIATION MEASURES

Based on the results from the hydraulic assessment, it has been decided to undertake remediation measures in order to reinforce the bank protection in Zone 3. The new bank protection is characterized as follows (see Fig. 6):

- Section 0+255 to 0+315: D_{50} = 0.90 m, which effectively corresponds to a prolongation of Zone 2 60 m inside Zone 3. Two layers of armourstone material (thickness 1.7 m) and two additional filter layers underneath.
- Section 0+315 to 0+435: D_{50} = 0.60 m. Two layers of armourstone material (thickness 1.2 m) and two additional filter layers underneath.

No remediation measures were defined downstream of Zone 3 as any possible damage in this region is not expected to affect dam safety.

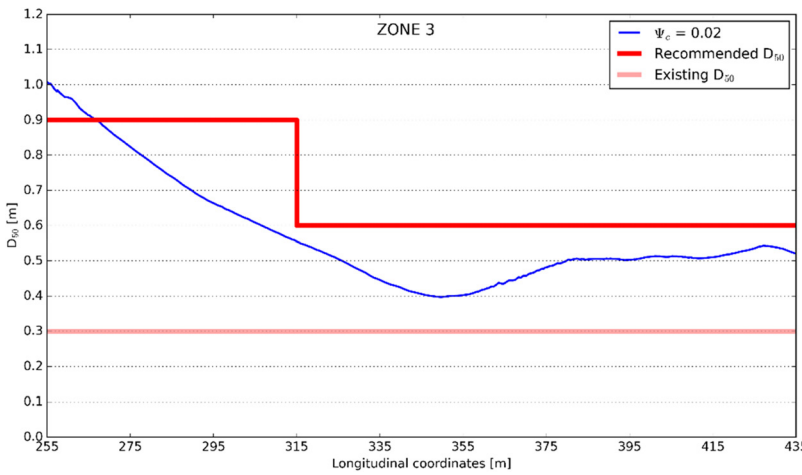


Fig. 6
Longitudinal profile along Zone 3 with existing and recommended D_{50} values. The blue line corresponds to the calculated D_{50} with Ψ_c = 0.02 (value chosen for bank protection design, see section 3.3) for the geometry with slumped material removed for the 100-year flood.
Profil en long de la Zone 3 avec D_{50} existant et valeurs recommandées. La courbe bleue matérialise le D_{50} calculé avec Ψ_c = 0.02 (valeur retenue pour le dimensionnement de la protection de berge, voir section 3.3) pour la configuration sans éboulement. Débit centennal.

5. CONCLUSIONS

In this article, a hydraulic assessment of flow conditions in a tailrace channel affected by bank protection damages has been presented. The hydraulic assessment has been performed with 2D hydraulic modelling combined with a turbulence-based bank protection design method in order to investigate the required armourstone size in the area affected by damages. Results from the 2D hydraulic model showed that high turbulence in the affected zone is likely to have facilitated the occurrence of damages. The armourstone size used in the existing bank protection does not fulfill the requirements from turbulence-based design methods.

Comparison between turbulence-based bank protection design methods and methods in which flow turbulence is not an explicit input parameter has shown that the latter yield potentially unconservative design values when high flow turbulence is expected. As a result, remediations measures have been defined in order to reinforce the bank protection in the region affected by damages (Zone 3). The new bank protection in this zone is characterized by armourstone material with median diameters D_{50} 0.90 m (section 0+255 to 0+315) and 0.60 m (section 0+315 to 0+435). It represents a significant upgrade compared with the existing armourstone material characterized by a median diameter D_{50} of 0.30 m.

The combination of multi-dimensional hydraulic and turbulence modelling combined with turbulence-based design methods can contribute to a better understanding of erosion phenomena and to optimal bank protection design.

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